

1	Course Name:
	DEEP LEARNING
2	Course Code:
	IQB20304
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	NA
6	Course Synopsis: This course comprehensively covers Deep Learning, blending foundational theory with hands-on application. Students will explore neural networks, CNNs, RNNs, transformers, GANs, and graph neural networks (GNNs), learning to design and optimize models for tasks like image classification, natural language processing, and reinforcement learning. Through labs using PyTorch and TensorFlow, learners will implement architectures (e.g., ResNet, BERT) and deploy solutions on edge devices. Ethical challenges – bias, explainability, environment impact – are critically examined, alongside model compression and deployment strategies. The course culminates in a capstone project solving real-world problems (e.g., healthcare, climate), equipping students with technical mastery and ethical responsibility for AI careers.
7	Course Learning Outcome: Upon completion of the course, students should be able to: 1. Analyze the architectural design, strengths, and limitations of deep learning models (e.g., CNNs, RNNs, Transformers, GANs) for solving real-world problems in computer vision, NLP, and reinforcement learning. 2. Build end-to-end deep learning systems (e.g., image classifiers, language models, reinforcement learning agents) using frameworks like PyTorch and TensorFlow, while optimizing for efficiency and scalability. 3. Propose mitigation strategies for deep learning systems in alignment with global standards.
8	Content Outline:
	Topic
	1. Foundations of Neural Networks
	2. Convolutional Neural Networks (CNNs)
	3. Recurrent Networks & Attention Mechanisms
	4. Advanced Deep Learning Models
	5. Deployment, Ethics, and Future Trends
9	References:

	Deep Learning: A Visual Approach" by Andrew Glassner (2025) (main reference). "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron (practical labs).
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